Supplementary material for Observe, Imagine, Abstain: Detecting Hallucinated Geometry in Feed-Forward 3D Reconstruction

Anonymous CVPR submission

Figure and table numbers below are local to this document; the main paper is self-contained and refers to this material by section title.

Compute. The benchmark involves no training, and all its model runs fit on one 16 GB GPU in about eight hours. The support head trains in ten to twenty-five minutes on one GPU (1500 steps); scoring one detector on one dataset takes one to five minutes on an H200. Every threshold and pass condition was fixed before the run that uses it.

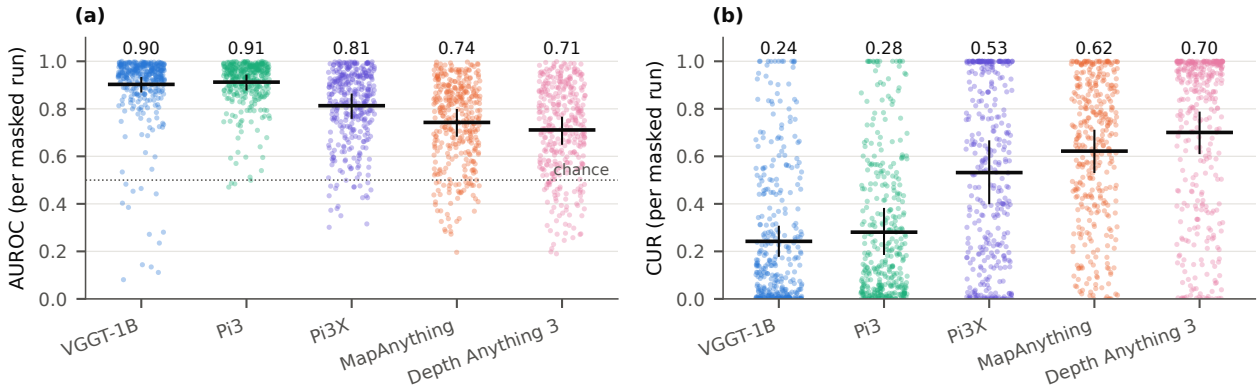


Figure S1. **Per-run spread on DTU.** AUROC and CUR for every (scan, seed, radius) run under blur and noise fills, with pooled means and intervals. The spread across scans is wide for every model; the intervals in the main tables come from resampling scans.

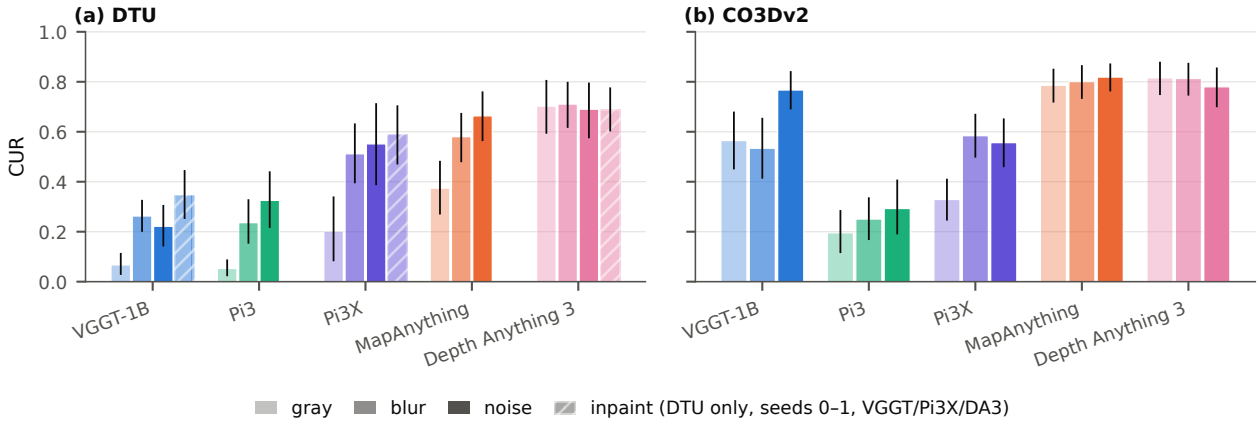


Figure S2. **Fill ablation.** CUR per model and fill on DTU and CO3Dv2 (five seeds) and for the inpainted fill on DTU (two seeds, three models). A visible gray patch gives the lowest leakage; natural-statistics and inpainted fills give more.

Correctness guards. Before any model is scored we check that ground-truth depth reprojects consistently between view pairs and that unsupported points of view i, where visible in view j, land inside view j's hidden mask. DTU: reprojection agreement 0.80 to 0.82 at a 1% depth tolerance, U-transfer 0.96 to 0.99. CO3Dv2 (cameras converted from PyTorch3D and checked against the official point cloud): with an occlusion-aware 3% tolerance, agreement 0.92 to 1.00 and U-transfer 0.90 to 0.98 on 17 of 18 scenes; the eighteenth scores 0.80, is kept, and removing it changes nothing (Section 4 of the paper). ETH3D: agreement 0.996 to 1.000 on all scenes; nine of twelve pass the U-transfer check and three are excluded before any model is run.

Pipeline validation. On a synthetic DTU-format scene, an oracle that knows what is hidden scores AUROC 1.000 and CUR 0.000, and a blind guesser with flat confidence scores AUROC 0.500 and CUR 0.90 (the 10% of supported pixels dropped by τ is the expected floor). These numbers validate the code, not any model.

Table 1 with intervals. The paper's Table 1 shows means; Table S1 adds 95% cluster-bootstrap intervals over scenes, CUR-ERR and the confidence ratio (median confidence inside the hidden region over its clean value), with the matched split's AUROC_m and CUR_m.

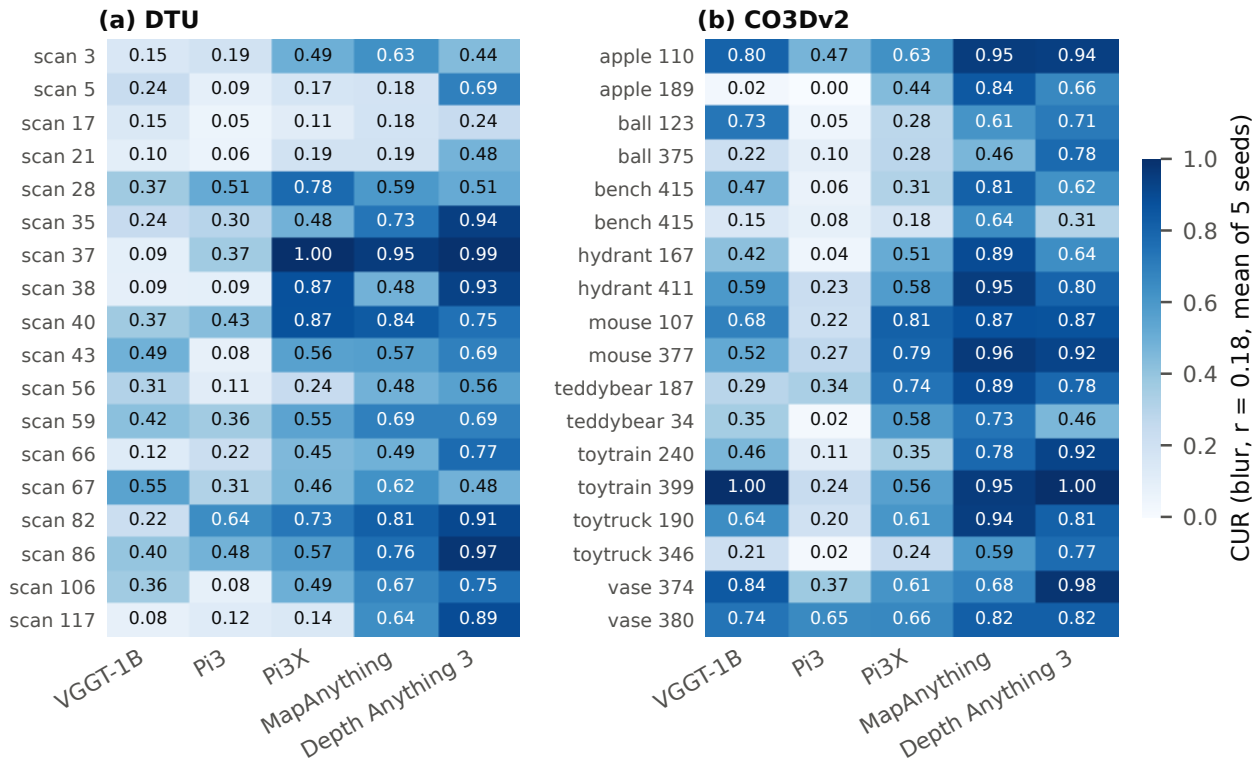


Figure S3. **Per-scene CUR** (blur fill, radius 0.18, mean over seeds) for every model on DTU scans and CO3Dv2 scenes. Rows are scenes, columns are models.

Table S1. **Native confidence, full numbers.** Blur and noise pooled, 18 scenes × 5 seeds × 2 radii; mean [95% CI]; matched split seeds 0 and 1, radius 0.18.

model	DTU, 18 scans						CO3Dv2, 18 scenes					
	AUROC↑	CUR↓	CUR-ERR↓	conf ratio	AUROC_m ↑	CUR_m ↓	AUROC↑	CUR↓	CUR-ERR↓	conf ratio	AUROC_m ↑	CUR_m ↓
VGGT-1B	0.903 [0.873, 0.929]	0.242 [0.181, 0.304]	0.000	0.22	0.551	0.802	0.681 [0.632, 0.729]	0.651 [0.566, 0.734]	0.031	0.70	0.486	0.852
Pi3	0.912 [0.881, 0.940]	0.281 [0.189, 0.379]	0.002	0.34	0.495	0.828	0.891 [0.845, 0.930]	0.272 [0.190, 0.360]	0.002	0.43	0.494	0.808
Pi3X	0.813 [0.762, 0.860]	0.532 [0.403, 0.663]	0.003	0.91	0.474	0.870	0.732 [0.680, 0.779]	0.570 [0.500, 0.641]	0.001	0.87	0.500	0.806
MapAnything	0.743 [0.687, 0.795]	0.622 [0.534, 0.707]	0.286	0.69	0.515	0.837	0.576 [0.521, 0.632]	0.810 [0.756, 0.863]	0.007	0.68	0.493	0.836
Depth Anything 3	0.711 [0.653, 0.763]	0.701 [0.614, 0.784]	0.002	0.83	0.425	0.854	0.620 [0.567, 0.670]	0.797 [0.728, 0.861]	0.000	0.88	0.482	0.839

Table S2. **DTU, five models, full configuration.** 18 scans × 5 mask seeds × 2 radii × 3 fills, pooled over radii; mean [95% cluster-bootstrap CI over scans]. *conf ratio* is the median confidence inside the hidden region divided by its value under clean input. The pooled numbers are in Table 1 of the main paper.

model	fill	AUROC	CUR	CUR-ERR	AbsRel_U	AbsRel_U ctrl	conf ratio
VGGT-1B	gray	0.973 [0.955, 0.987]	0.067 [0.031, 0.111]	0.000	0.018	0.006	0.051
VGGT-1B	blur	0.886 [0.843, 0.922]	0.263 [0.203, 0.324]	0.000	0.011	0.006	0.241
VGGT-1B	noise	0.919 [0.892, 0.945]	0.222 [0.145, 0.304]	0.000	0.014	0.006	0.192
Pi3	gray	0.981 [0.972, 0.988]	0.053 [0.025, 0.085]	0.001	0.020	0.014	0.057
Pi3	blur	0.923 [0.895, 0.947]	0.236 [0.156, 0.326]	0.001	0.018	0.014	0.261
Pi3	noise	0.902 [0.865, 0.934]	0.325 [0.219, 0.438]	0.002	0.020	0.014	0.411
Pi3X	gray	0.940 [0.902, 0.971]	0.202 [0.085, 0.337]	0.002	0.022	0.017	0.335
Pi3X	blur	0.826 [0.782, 0.865]	0.512 [0.397, 0.630]	0.002	0.020	0.017	0.739
Pi3X	noise	0.801 [0.734, 0.864]	0.551 [0.390, 0.711]	0.004	0.022	0.017	1.091
MapAnything	gray	0.861 [0.812, 0.905]	0.374 [0.272, 0.480]	0.195 [0.125, 0.269]	0.222	0.111	0.650
MapAnything	blur	0.773 [0.720, 0.825]	0.580 [0.482, 0.672]	0.277 [0.197, 0.361]	0.135	0.111	0.677
MapAnything	noise	0.712 [0.645, 0.773]	0.664 [0.566, 0.758]	0.295 [0.211, 0.390]	0.129	0.111	0.709
Depth Anything 3	gray	0.723 [0.664, 0.780]	0.703 [0.596, 0.804]	0.003	0.017	0.013	0.804
Depth Anything 3	blur	0.682 [0.611, 0.744]	0.711 [0.619, 0.796]	0.003	0.016	0.013	0.800
Depth Anything 3	noise	0.740 [0.686, 0.793]	0.690 [0.578, 0.793]	0.001	0.017	0.013	0.855

DTU and CO3Dv2, full configuration per fill.

Table S3. **CO3Dv2, five models, full configuration.** 18 scenes × 5 seeds × 2 radii × 3 fills, pooled over radii; mean [95% CI over scenes]. The pooled numbers are quoted in Section 5.3 of the main paper.

model	fill	AUROC	CUR	CUR-ERR	AbsRel$_S$	AbsRel$_U$	AbsRel$_U$ ctrl	conf ratio
VGGT-1B	gray	0.747 [0.692, 0.802]	0.565 [0.453, 0.677]	0.029	0.029	0.031	0.027	0.634
VGGT-1B	blur	0.772 [0.717, 0.825]	0.534 [0.416, 0.652]	0.034	0.029	0.031	0.027	0.603
VGGT-1B	noise	0.590 [0.517, 0.656]	0.767 [0.693, 0.839]	0.029	0.029	0.029	0.027	0.805
Pi3	gray	0.930 [0.894, 0.958]	0.196 [0.118, 0.283]	0.002	0.015	0.016	0.013	0.351
Pi3	blur	0.903 [0.860, 0.939]	0.251 [0.170, 0.334]	0.002	0.015	0.016	0.013	0.389
Pi3	noise	0.880 [0.813, 0.929]	0.293 [0.193, 0.405]	0.002	0.015	0.015	0.013	0.472
Pi3X	gray	0.863 [0.825, 0.899]	0.329 [0.248, 0.408]	0.001	0.017	0.022	0.014	0.749
Pi3X	blur	0.720 [0.661, 0.777]	0.584 [0.500, 0.668]	0.002	0.017	0.018	0.014	0.877
Pi3X	noise	0.743 [0.680, 0.798]	0.556 [0.462, 0.650]	0.001	0.017	0.019	0.014	0.867
MapAnything	gray	0.609 [0.554, 0.665]	0.786 [0.721, 0.849]	0.004	0.041	0.029	0.034	0.632
MapAnything	blur	0.575 [0.514, 0.637]	0.801 [0.735, 0.863]	0.008	0.042	0.028	0.034	0.674
MapAnything	noise	0.578 [0.524, 0.631]	0.819 [0.765, 0.870]	0.007	0.041	0.033	0.034	0.690
Depth Anything 3	gray	0.619 [0.560, 0.674]	0.816 [0.751, 0.877]	0.001	0.019	0.015	0.014	0.874
Depth Anything 3	blur	0.622 [0.574, 0.668]	0.813 [0.748, 0.872]	0.001	0.018	0.015	0.014	0.853
Depth Anything 3	noise	0.618 [0.553, 0.679]	0.780 [0.702, 0.854]	0.000	0.019	0.016	0.014	0.901

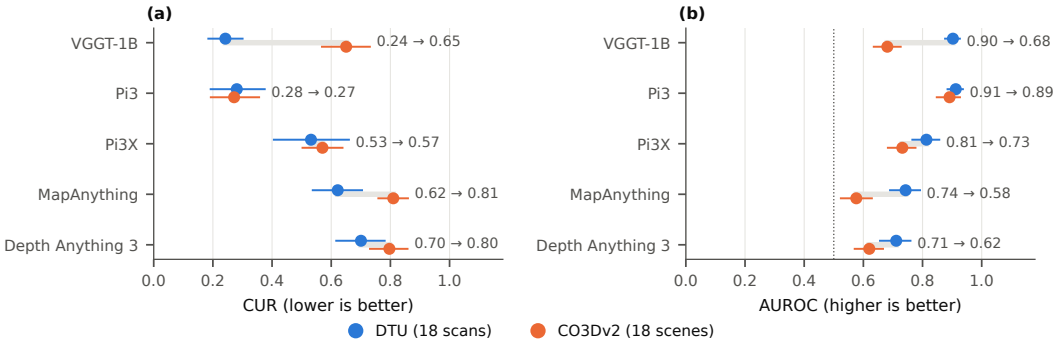


Figure S4. **Lab rig versus phone captures.** Pooled blur-and-noise CUR and AUROC per model on DTU and CO3Dv2 with 95% intervals. Pi3 is the only model whose numbers transfer; VGGT, MapAnything and DA3 leak markedly more on CO3Dv2.

Lab rig versus phone captures, plotted.

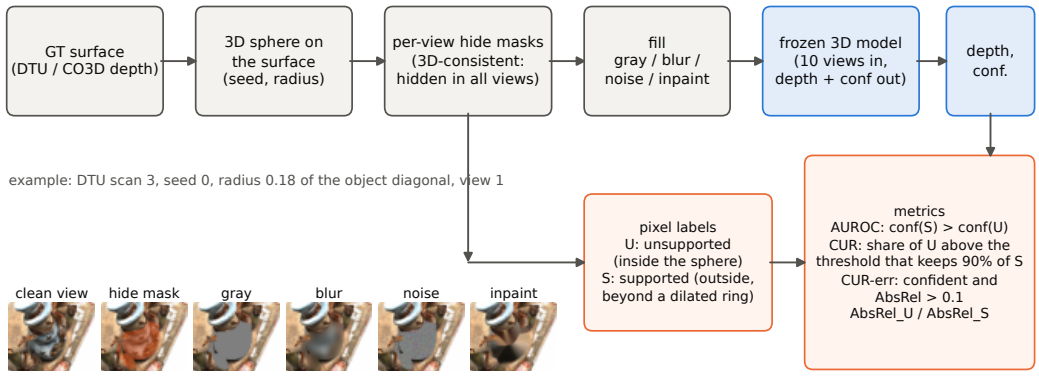


Figure S5. **The benchmark protocol.** A sphere is placed on the ground-truth surface; every pixel in every view whose ground-truth point falls inside it is hidden, so the hidden region is 3D-consistent. Hidden pixels are filled (gray, blur, noise or inpaint), the model runs on the filled views, and its depth and confidence are scored against the unsupported (U), supported (S) and ring labels. The same pixels are also scored under clean input as a control.

The benchmark protocol, in a figure.

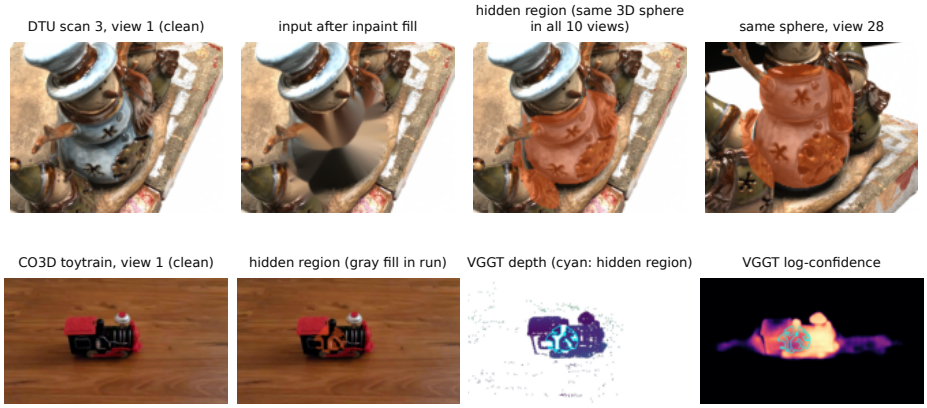


Figure S6. **What a model does with a surface it cannot see.** Top: a DTU scan with one 3D region of the ground-truth surface hidden in all ten input views and inpainted so no seam remains (the inpainted fill of Section 4 of the paper). Bottom: a CO3Dv2 phone capture treated the same way. The top row shows the protocol: the same 3D sphere hides the region in every view, including a view from the other side of the rig. The bottom row shows what the backbone (VGGT) does with it: plausible geometry inside the hidden region, with confidence close to its value outside. The cyan contour marks the unsupported pixels. Model panels are the recorded run outputs.

The filled patch, in pictures.

Fused point clouds at matched coverage. Each backbone's per-view depths are back-projected with its own predicted cameras into one cloud, aligned to the ground-truth frame by a similarity transform fitted on the camera centres, and compared with the ground-truth cloud of the same views at a tolerance of 1% of the scene's median depth: accuracy is the fraction of predicted points within tolerance of a ground-truth point, completeness the converse. A detector keeps the given fraction of points with the highest score; random dropping and an oracle that drops single-view pixels first are the references. Table S4: 18 DTU scans and 9 ETH3D scenes, 5 and 10 views, two view sets; the rows marked $\times$ are the label-free head multiplied by native confidence. The absolute accuracies on ETH3D are low because the predicted cameras of a building scan align poorly to the laser frame; the comparison between detectors at matched coverage is what the table is for.

Table S4. **Fused point cloud at matched coverage**, accuracy / completeness against the laser or MVS cloud.

detector	DTU, accuracy / completeness			ETH3D, accuracy / completeness		
coverage	0.6	0.8	1.0	0.6	0.8	1.0
native confidence	0.842 / 0.804	0.783 / 0.885	0.653 / 0.902	0.337 / 0.396	0.306 / 0.443	0.273 / 0.470
reprojection consistency	0.805 / 0.837	0.763 / 0.868	0.654 / 0.902	0.321 / 0.413	0.302 / 0.438	0.273 / 0.470
per-image control	0.766 / 0.735	0.698 / 0.835	0.653 / 0.902	0.293 / 0.369	0.281 / 0.428	0.273 / 0.470
label-free head	0.834 / 0.770	0.771 / 0.874	0.653 / 0.902	0.331 / 0.383	0.300 / 0.428	0.273 / 0.470
random dropping	0.653 / 0.896	0.653 / 0.900	0.653 / 0.902	0.273 / 0.462	0.273 / 0.467	0.273 / 0.470
oracle: single-view first	0.699 / 0.833	0.663 / 0.863	0.653 / 0.902	0.273 / 0.417	0.275 / 0.442	0.273 / 0.470
label-free head $\times$ native confidence	0.844 / 0.786	0.785 / 0.883	0.653 / 0.902	0.333 / 0.384	0.304 / 0.437	0.273 / 0.470
native confidence (Pi3X)	0.381 / 0.498	0.349 / 0.543	0.297 / 0.562	0.346 / 0.426	0.314 / 0.467	0.272 / 0.482
reprojection (Pi3X)	0.353 / 0.485	0.350 / 0.535	0.297 / 0.559	0.301 / 0.412	0.297 / 0.439	0.273 / 0.481
label-free head (Pi3X)	0.404 / 0.494	0.354 / 0.546	0.297 / 0.562	0.332 / 0.404	0.299 / 0.446	0.272 / 0.482
label-free head $\times$ native (Pi3X)	0.403 / 0.492	0.356 / 0.547	0.297 / 0.562	0.338 / 0.408	0.306 / 0.454	0.272 / 0.481
oracle (Pi3X)	0.310 / 0.500	0.300 / 0.531	0.297 / 0.562	0.260 / 0.418	0.268 / 0.449	0.272 / 0.482

Detectors on the filled patches. The cheap alternatives and the fill-trained heads of Sections 4 and 6 of the main paper, on the Table 1 rows with blur, radius 0.18 and seed 0. A fuller perturbation baseline (photometric, blur, flip and view-permutation perturbations, $K = 8$, least-squares scale alignment, median-absolute-deviation spread) scores AUROC 0.52 to 0.60 and CUR 0.74 to 0.86 on all five backbones over blur and noise, worse than native confidence everywhere. On 22 further DTU scans (the MVSNet test split, seed 0, radius 0.18) the ranking of Table 1 holds: VGGT AUROC 0.908 / CUR 0.233, Pi3 0.919 / 0.272, Pi3X 0.850 / 0.539, DA3 0.755 / 0.650, MapAnything 0.697 / 0.693 with CUR-ERR 0.287. On the nine ETH3D scenes

under the full fill protocol native confidence leaks more than on DTU, CUR 0.61 (Pi3) to 0.79 (VGGT), and the fill-trained head scores 0.981 against its per-image control's 0.987.

Table S5. **Detectors on the filled patches, DTU** (18 scans, seed 0, blur, radius 0.18; native and perturbation rows re-run on the detectors' hardware, so every pair is scored on identical rows). The three learned rows per backbone are the *fill-trained* heads: they find the fill, not the missing support (Section 5 of the paper), which is why the per-image control matches them. The visibility-trained heads are in the main paper.

backbone	detector	fwd	AUROC	CUR
VGGT-1B	native confidence	1	0.915 [0.883, 0.943]	0.255 [0.157, 0.364]
VGGT-1B	perturbation var., $K{=}4$	4	0.666 [0.602, 0.729]	0.627 [0.521, 0.732]
VGGT-1B	reprojection consistency	1	0.488 [0.473, 0.506]	0.931 [0.903, 0.954]
VGGT-1B	fill-trained linear probe	1	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
VGGT-1B	fill-trained per-image control	1	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
VGGT-1B	fill-trained support head	1	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
Pi3X	native confidence	1	0.832 [0.782, 0.879]	0.502 [0.362, 0.642]
Pi3X	perturbation var., $K{=}4$	4	0.550 [0.491, 0.608]	0.824 [0.754, 0.886]
Pi3X	reprojection consistency	1	0.478 [0.435, 0.518]	0.929 [0.899, 0.955]
Pi3X	fill-trained per-image control	1	1.000 [0.999, 1.000]	0.000 [0.000, 0.001]
Pi3X	fill-trained support head	1	0.999 [0.999, 1.000]	0.001 [0.000, 0.003]

Stress tests. Section 5.2 of the main paper summarises this table.

Table S6. **Stress tests on DTU.** 18 scans $\times$ 2 mask seeds, radius 0.18, models VGGT, Pi3X, DA3. Reference rows are the matching cells of Table 1 of the paper (blur, 10 spread views, seeds 0 and 1). Last column: paired CUR difference against the reference with a sign-flip p over scans.

model	test	AUROC	CUR	CUR-ERR	AbsRel_U	CUR vs ref
VGGT-1B	reference	0.902 [0.868, 0.930]	0.259 [0.175, 0.349]	0.000	0.010	
VGGT-1B	inpaint fill	0.867 [0.821, 0.907]	0.347 [0.255, 0.443]	0.001	0.018	$+0.088$ ($p = 0.031$)
VGGT-1B	5 views	0.902	0.269 [0.173, 0.377]	0.000	0.011	$+0.010$ ($p = 0.63$)
VGGT-1B	3 views	0.899	0.259 [0.167, 0.361]	0.000	0.013	$+0.001$ ($p = 0.98$)
VGGT-1B	low parallax	0.911	0.214 [0.110, 0.334]	0.000	0.021	-0.045 ($p = 0.35$)
VGGT-1B	+ distractor	0.905	0.257 [0.179, 0.340]	0.000	0.011	-0.002 ($p = 0.90$)
Pi3X	reference	0.828 [0.781, 0.870]	0.521 [0.390, 0.650]	0.002	0.024	
Pi3X	inpaint fill	0.786 [0.731, 0.836]	0.589 [0.473, 0.702]	0.004	0.022	$+0.068$ ($p = 0.005$)
Pi3X	5 views	0.822	0.539 [0.411, 0.661]	0.006	0.024	$+0.018$ ($p = 0.44$)
Pi3X	3 views	0.840	0.497 [0.373, 0.618]	0.032	0.051	-0.024 ($p = 0.64$)
Pi3X	low parallax	0.812	0.523 [0.376, 0.671]	0.040	0.050	$+0.002$ ($p = 0.98$)
Pi3X	+ distractor	0.829	0.510 [0.368, 0.650]	0.002	0.024	-0.011 ($p = 0.66$)
DA3	reference	0.712 [0.632, 0.783]	0.679 [0.569, 0.791]	0.006	0.018	
DA3	inpaint fill	0.726	0.688 [0.605, 0.774]	0.009	0.018	$+0.008$ ($p = 0.87$)
DA3	5 views	0.697	0.785 [0.703, 0.865]	0.005	0.019	$+0.106$ ($p = 0.008$)
DA3	3 views	0.729	0.667 [0.560, 0.770]	0.018	0.031	-0.012 ($p = 0.81$)
DA3	low parallax	0.863 [0.785, 0.925]	0.372 [0.230, 0.520]	0.008	0.029	-0.307 ($p = 0.0003$)
DA3	+ distractor	0.660 [0.580, 0.734]	0.775 [0.680, 0.868]	0.003	0.019	$+0.096$ ($p = 0.0004$)

Stress tests, plotted.

Every detector on the true-visibility protocol. Table S7 lists every detector of the main paper's Table 3 with 95% cluster-bootstrap intervals over scenes, including the fill-trained heads and the linear probe trained on self-support labels.

Operating points from the per-pixel scores. Table S8 is computed from the per-pixel scores on the labelled pixels of every cell (scene, seed, view count), each cell weighing the same. AURC: area under the single-view rate of the kept pixels against coverage (oracle: supported pixels first). AbsRel of all labelled pixels and of the half with the highest score. CUR / keep: fraction of single-view pixels passing and fraction of supported pixels kept at a threshold fixed (a) on the other scenes of the same dataset so that 90% of their supported pixels pass (leave-one-scene-out), (b) on DTU and applied unchanged (frozen; the DTU row has none). Bands: fraction of pixels marked observed / abstain / imagined at the paper's fixed thresholds; the single-view rate and AbsRel inside the observed band.

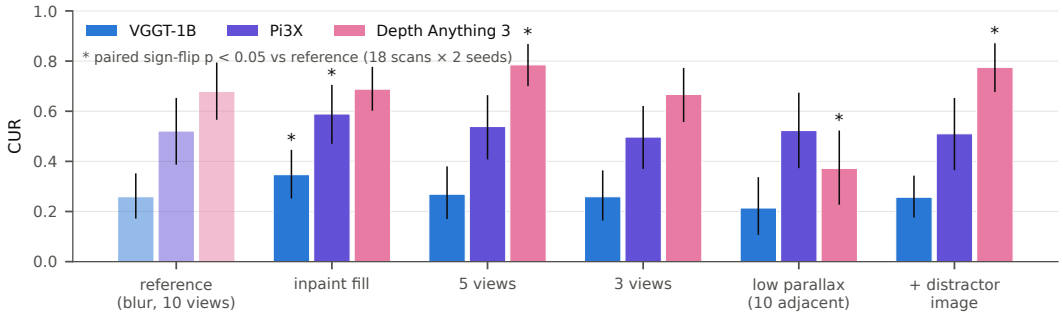


Figure S7. **Stress tests.** CUR under each stress condition against the reference, with intervals. View count does nothing for VGGT and Pi3X; the inpainted fill raises leakage; DA3 alone reacts to parallax and to a distractor.

Table S7. **True-visibility protocol, every detector**, pooled over scenes and view counts as in the main paper, with 95% cluster-bootstrap intervals over scenes.

detector	DTU AUROC↑	DTU CUR↓	CO3Dv2 AUROC↑	CO3Dv2 CUR↓	ETH3D AUROC↑	ETH3D CUR↓	Photot. AUROC↑	Photot. CUR↓
VGGT-1B								
native confidence	0.772 [0.722, 0.823]	0.530 [0.443, 0.616]	0.754 [0.712, 0.793]	0.723 [0.650, 0.796]	0.715 [0.621, 0.802]	0.566 [0.430, 0.704]	0.802 [0.748, 0.850]	0.451 [0.343, 0.569]
reprojection consistency	0.954 [0.944, 0.963]	0.082 [0.067, 0.098]	0.793 [0.746, 0.839]	0.693 [0.517, 0.860]	0.975 [0.968, 0.982]	0.069 [0.034, 0.106]	0.824 [0.786, 0.869]	0.340 [0.249, 0.423]
distance to image border	0.701 [0.671, 0.731]	0.727 [0.700, 0.755]	0.454 [0.388, 0.520]	0.897 [0.834, 0.946]	0.725 [0.669, 0.780]	0.632 [0.534, 0.728]	0.730 [0.695, 0.765]	0.649 [0.606, 0.689]
depth edges	0.549 [0.519, 0.576]	0.831 [0.806, 0.855]	0.422 [0.373, 0.473]	0.943 [0.923, 0.962]	0.598 [0.503, 0.699]	0.762 [0.638, 0.857]	0.674 [0.594, 0.748]	0.619 [0.527, 0.714]
fill-trained head	0.609 [0.575, 0.646]	0.765 [0.728, 0.798]	0.451 [0.379, 0.522]	0.916 [0.880, 0.948]	0.686 [0.615, 0.760]	0.657 [0.551, 0.758]	0.704 [0.647, 0.759]	0.673 [0.594, 0.749]
fill-trained per-image control	0.744 [0.705, 0.779]	0.640 [0.586, 0.697]	0.356 [0.306, 0.410]	0.957 [0.943, 0.969]	0.619 [0.542, 0.698]	0.736 [0.641, 0.820]	0.622 [0.562, 0.686]	0.760 [0.706, 0.805]
per-image control (GT labels)	0.878 [0.845, 0.907]	0.340 [0.289, 0.394]	0.462 [0.392, 0.540]	0.878 [0.814, 0.933]	0.712 [0.649, 0.767]	0.666 [0.608, 0.731]	0.787 [0.732, 0.837]	0.530 [0.448, 0.603]
linear probe (GT labels)	0.989 [0.985, 0.992]	0.024 [0.015, 0.035]	0.913 [0.883, 0.942]	0.250 [0.165, 0.337]	0.966 [0.959, 0.975]	0.095 [0.068, 0.122]	0.894 [0.843, 0.936]	0.265 [0.158, 0.392]
support head, GT visibility labels	0.993 [0.990, 0.995]	0.015 [0.008, 0.024]	0.927 [0.903, 0.951]	0.211 [0.138, 0.287]	0.980 [0.972, 0.986]	0.055 [0.035, 0.076]	0.908 [0.863, 0.947]	0.258 [0.146, 0.391]
per-image control (self labels)	0.885 [0.851, 0.912]	0.325 [0.272, 0.383]	0.460 [0.394, 0.532]	0.892 [0.833, 0.937]	0.626 [0.538, 0.711]	0.750 [0.665, 0.830]	0.729 [0.656, 0.805]	0.681 [0.535, 0.810]
linear probe (self labels)	0.988 [0.984, 0.992]	0.022 [0.014, 0.032]	0.898 [0.861, 0.934]	0.296 [0.191, 0.402]	0.967 [0.958, 0.976]	0.084 [0.058, 0.107]	0.924 [0.903, 0.948]	0.187 [0.127, 0.243]
support head, label-free	0.992 [0.989, 0.995]	0.017 [0.010, 0.025]	0.931 [0.907, 0.953]	0.199 [0.134, 0.266]	0.980 [0.972, 0.986]	0.051 [0.033, 0.069]	0.930 [0.908, 0.954]	0.191 [0.125, 0.253]
label-free head, adapted on the target domain	–	–	0.841 [0.785, 0.891]	0.474 [0.338, 0.610]	–	–	0.926 [0.899, 0.952]	0.203 [0.132, 0.275]
support head (GT labels) × native confidence	0.986 [0.980, 0.990]	0.031 [0.018, 0.047]	0.920 [0.891, 0.947]	0.236 [0.149, 0.338]	0.954 [0.935, 0.972]	0.131 [0.071, 0.199]	0.931 [0.912, 0.952]	0.177 [0.115, 0.237]
label-free head × native confidence	0.980 [0.973, 0.986]	0.048 [0.030, 0.069]	0.920 [0.890, 0.948]	0.225 [0.144, 0.314]	0.943 [0.920, 0.965]	0.150 [0.086, 0.224]	0.930 [0.912, 0.951]	0.179 [0.118, 0.238]
Pi3X								
native confidence	0.820 [0.793, 0.845]	0.493 [0.435, 0.553]	0.697 [0.649, 0.740]	0.846 [0.803, 0.888]	0.887 [0.856, 0.918]	0.304 [0.213, 0.389]	0.839 [0.808, 0.867]	0.450 [0.365, 0.549]
reprojection consistency	0.914 [0.900, 0.929]	0.164 [0.120, 0.213]	0.885 [0.859, 0.909]	0.224 [0.164, 0.293]	0.977 [0.965, 0.986]	0.054 [0.024, 0.092]	0.814 [0.775, 0.857]	0.361 [0.287, 0.431]
distance to image border	0.701 [0.671, 0.731]	0.727 [0.700, 0.755]	0.454 [0.388, 0.520]	0.897 [0.834, 0.946]	0.725 [0.669, 0.780]	0.632 [0.534, 0.728]	0.730 [0.695, 0.765]	0.649 [0.606, 0.689]
depth edges	0.545 [0.516, 0.573]	0.835 [0.812, 0.858]	0.424 [0.377, 0.474]	0.951 [0.933, 0.967]	0.596 [0.498, 0.703]	0.761 [0.640, 0.854]	0.638 [0.570, 0.700]	0.699 [0.640, 0.762]
fill-trained head	0.713 [0.679, 0.745]	0.650 [0.596, 0.706]	0.433 [0.378, 0.491]	0.909 [0.864, 0.942]	0.647 [0.573, 0.703]	0.695 [0.643, 0.751]	0.823 [0.784, 0.863]	0.454 [0.370, 0.536]
fill-trained per-image control	0.718 [0.681, 0.751]	0.667 [0.622, 0.715]	0.391 [0.338, 0.445]	0.959 [0.948, 0.969]	0.644 [0.595, 0.695]	0.719 [0.625, 0.800]	0.681 [0.631, 0.730]	0.720 [0.657, 0.781]
per-image control (GT labels)	0.863 [0.831, 0.891]	0.375 [0.326, 0.425]	0.428 [0.359, 0.507]	0.889 [0.823, 0.941]	0.694 [0.640, 0.749]	0.685 [0.614, 0.745]	0.775 [0.722, 0.825]	0.577 [0.499, 0.651]
linear probe (GT labels)	0.985 [0.979, 0.989]	0.034 [0.021, 0.049]	0.931 [0.907, 0.954]	0.203 [0.133, 0.277]	0.956 [0.939, 0.970]	0.123 [0.074, 0.181]	0.919 [0.894, 0.944]	0.204 [0.136, 0.271]
support head, GT visibility labels	0.990 [0.986, 0.994]	0.021 [0.010, 0.035]	0.930 [0.907, 0.953]	0.221 [0.138, 0.312]	0.975 [0.964, 0.984]	0.072 [0.041, 0.107]	0.938 [0.919, 0.959]	0.176 [0.111, 0.240]
per-image control (self labels)	0.871 [0.837, 0.899]	0.363 [0.309, 0.418]	0.469 [0.400, 0.546]	0.877 [0.809, 0.928]	0.610 [0.531, 0.688]	0.776 [0.709, 0.831]	0.719 [0.643, 0.797]	0.697 [0.566, 0.821]
linear probe (self labels)	0.983 [0.977, 0.988]	0.036 [0.023, 0.051]	0.931 [0.903, 0.957]	0.192 [0.121, 0.269]	0.939 [0.918, 0.956]	0.171 [0.114, 0.241]	0.919 [0.894, 0.945]	0.200 [0.133, 0.264]
support head, label-free	0.988 [0.983, 0.993]	0.025 [0.013, 0.040]	0.939 [0.916, 0.960]	0.190 [0.118, 0.265]	0.963 [0.946, 0.976]	0.103 [0.061, 0.158]	0.940 [0.920, 0.960]	0.166 [0.105, 0.226]
label-free head, adapted on the target domain	–	–	0.964 [0.947, 0.978]	0.094 [0.054, 0.143]	–	–	0.929 [0.898, 0.958]	0.198 [0.118, 0.282]
support head (GT labels) × native confidence	0.988 [0.983, 0.993]	0.027 [0.014, 0.042]	0.928 [0.903, 0.952]	0.230 [0.144, 0.323]	0.978 [0.967, 0.987]	0.064 [0.033, 0.099]	0.935 [0.911, 0.957]	0.181 [0.109, 0.258]
label-free head × native confidence	0.986 [0.980, 0.991]	0.030 [0.017, 0.046]	0.935 [0.912, 0.958]	0.200 [0.124, 0.280]	0.967 [0.949, 0.980]	0.092 [0.051, 0.146]	0.934 [0.910, 0.957]	0.185 [0.113, 0.258]
Pi3								
native confidence	0.841 [0.815, 0.864]	0.489 [0.426, 0.553]	0.551 [0.503, 0.600]	0.882 [0.845, 0.917]	0.814 [0.756, 0.871]	0.498 [0.371, 0.618]	0.847 [0.795, 0.892]	0.447 [0.345, 0.555]
reprojection consistency	0.952 [0.943, 0.960]	0.089 [0.069, 0.112]	0.894 [0.871, 0.915]	0.205 [0.153, 0.261]	0.976 [0.965, 0.985]	0.039 [0.023, 0.058]	0.817 [0.775, 0.864]	0.345 [0.250, 0.432]
distance to image border	0.701 [0.671, 0.731]	0.727 [0.700, 0.755]	0.454 [0.388, 0.520]	0.897 [0.834, 0.946]	0.725 [0.669, 0.780]	0.632 [0.534, 0.728]	0.730 [0.695, 0.765]	0.649 [0.606, 0.689]
depth edges	0.546 [0.518, 0.574]	0.839 [0.816, 0.862]	0.424 [0.377, 0.472]	0.950 [0.932, 0.967]	0.597 [0.499, 0.704]	0.764 [0.637, 0.858]	0.652 [0.583, 0.718]	0.670 [0.598, 0.745]
per-image control (GT labels)	0.878 [0.846, 0.906]	0.345 [0.291, 0.402]	0.464 [0.393, 0.544]	0.880 [0.815, 0.935]	0.717 [0.651, 0.774]	0.661 [0.601, 0.729]	0.796 [0.744, 0.842]	0.510 [0.431, 0.583]
linear probe (GT labels)	0.985 [0.980, 0.990]	0.032 [0.019, 0.046]	0.943 [0.922, 0.962]	0.175 [0.109, 0.245]	0.938 [0.905, 0.965]	0.165 [0.091, 0.249]	0.914 [0.882, 0.944]	0.218 [0.143, 0.290]
support head, GT visibility labels	0.992 [0.988, 0.995]	0.016 [0.008, 0.026]	0.934 [0.914, 0.954]	0.203 [0.135, 0.273]	0.953 [0.926, 0.974]	0.135 [0.068, 0.214]	0.932 [0.911, 0.954]	0.201 [0.131, 0.272]
per-image control (self labels)	0.888 [0.856, 0.914]	0.320 [0.268, 0.381]	0.471 [0.402, 0.548]	0.895 [0.834, 0.942]	0.654 [0.579, 0.725]	0.750 [0.668, 0.822]	0.724 [0.645, 0.804]	0.683 [0.533, 0.820]
linear probe (self labels)	0.983 [0.977, 0.989]	0.031 [0.018, 0.046]	0.926 [0.897, 0.953]	0.235 [0.139, 0.335]	0.917 [0.885, 0.944]	0.225 [0.151, 0.308]	0.915 [0.882, 0.945]	0.209 [0.140, 0.275]
support head, label-free	0.989 [0.984, 0.994]	0.021 [0.011, 0.034]	0.930 [0.907, 0.952]	0.225 [0.145, 0.309]	0.951 [0.924, 0.971]	0.139 [0.075, 0.215]	0.931 [0.908, 0.955]	0.181 [0.121, 0.238]
label-free head × native confidence	0.985 [0.979, 0.990]	0.034 [0.021, 0.048]	0.915 [0.888, 0.942]	0.266 [0.172, 0.364]	0.943 [0.911, 0.966]	0.151 [0.088, 0.228]	0.929 [0.907, 0.953]	0.197 [0.129, 0.257]
Depth Anything 3								
native confidence	0.836 [0.815, 0.855]	0.430 [0.383, 0.481]	0.683 [0.615, 0.742]	0.743 [0.655, 0.828]	0.643 [0.557, 0.744]	0.635 [0.494, 0.757]	0.782 [0.707, 0.851]	0.512 [0.389, 0.636]
reprojection consistency	0.926 [0.912, 0.939]	0.175 [0.118, 0.241]	0.836 [0.799, 0.872]	0.463 [0.335, 0.585]	0.962 [0.942, 0.977]	0.109 [0.044, 0.176]	0.844 [0.804, 0.882]	0.332 [0.247, 0.419]
distance to image border	0.701 [0.671, 0.731]	0.727 [0.700, 0.755]	0.454 [0.388, 0.520]	0.897 [0.834, 0.946]	0.725 [0.669, 0.780]	0.632 [0.534, 0.728]	0.730 [0.695, 0.765]	0.649 [0.606, 0.689]
depth edges	0.551 [0.525, 0.576]	0.844 [0.821, 0.868]	0.408 [0.356, 0.463]	0.968 [0.953, 0.981]	0.592 [0.498, 0.695]	0.767 [0.643, 0.859]	0.645 [0.582, 0.707]	0.717 [0.651, 0.782]
per-image control (GT labels)	0.761 [0.728, 0.794]	0.586 [0.537, 0.632]	0.440 [0.373, 0.506]	0.903 [0.842, 0.947]	0.746 [0.701, 0.794]	0.620 [0.535, 0.700]	0.732 [0.688, 0.767]	0.646 [0.606, 0.684]
linear probe (GT labels)	0.973 [0.964, 0.980]	0.069 [0.048, 0.093]	0.875 [0.846, 0.902]	0.354 [0.283, 0.427]	0.969 [0.956, 0.980]	0.086 [0.050, 0.125]	0.909 [0.884, 0.937]	0.216 [0.153, 0.270]
support head, GT visibility labels	0.985 [0.978, 0.990]	0.036 [0.022, 0.053]	0.883 [0.859, 0.908]	0.329 [0.265, 0.392]	0.985 [0.981, 0.989]	0.040 [0.026, 0.054]	0.927 [0.900, 0.953]	0.188 [0.127, 0.244]
per-image control (self labels)	0.760 [0.727, 0.792]	0.598 [0.548, 0.647]	0.416 [0.349, 0.484]	0.911 [0.847, 0.956]	0.749 [0.700, 0.801]	0.610 [0.514, 0.700]	0.727 [0.683, 0.764]	0.650 [0.608, 0.691]
linear probe (self labels)	0.971 [0.961, 0.978]	0.074 [0.050, 0.102]	0.869 [0.836, 0.900]	0.371 [0.292, 0.451]	0.952 [0.933, 0.967]	0.134 [0.089, 0.189]	0.909 [0.886, 0.937]	0.230 [0.158, 0.284]
support head, label-free	0.982 [0.975, 0.988]	0.040 [0.026, 0.058]	0.879 [0.844, 0.911]	0.323 [0.249, 0.398]	0.974 [0.968, 0.980]	0.070 [0.049, 0.092]	0.925 [0.905, 0.948]	0.202 [0.141, 0.250]
label-free head × native confidence	0.977 [0.967, 0.984]	0.056 [0.033, 0.083]	0.833 [0.784, 0.875]	0.394 [0.302, 0.490]	0.919 [0.892, 0.947]	0.167 [0.115, 0.220]	0.921 [0.898, 0.948]	0.206 [0.139, 0.264]

Cross-model disagreement. Two backbones run on the same views; view by view the second depth is aligned to the first by the median ratio over all valid pixels, and minus the relative disagreement is the confidence. Table S9: on the benchmark's filled patches every pair is at chance, and on real single-view geometry (DTU, clean views) disagreement scores 0.50 to 0.63, so models also agree with one another where only one view supports the surface. Agreement between models, unlike agreement between views, carries no signal in either case.

Fill sensitivity of the invented surface. If the surface a backbone returns inside the hidden region were fixed by the prior, changing the pixels the region is filled with would not move it. For every benchmark cell (seed 0, radius 0.18) the same hidden sphere is filled three ways (gray, blur, noise), the backbone runs on each, the three depths are aligned to the blur run

Table S8. **Operating points.** See the paragraph above for the columns.

dataset	backbone	detector	AURC	AbsRel all	AbsRel top half	CUR / keep (a)	CUR / keep (b)	obs. / abst. / imag.	single-view in obs.	AbsRel in obs.
DTU	VGGT-1B	native confidence	0.077	0.007	0.004	0.63 / 0.89	–	–	–	–
DTU	VGGT-1B	reprojection	0.035	0.007	0.005	0.08 / 0.94	–	–	–	–
DTU	VGGT-1B	head, GT labels	0.025	0.007	0.004	0.01 / 0.90	–	0.82 / 0.04 / 0.15	0.007	0.005
DTU	VGGT-1B	head, label-free	0.026	0.007	0.004	0.01 / 0.90	–	0.82 / 0.03 / 0.14	0.008	0.005
DTU	Pi3X	native confidence	0.057	0.021	0.015	0.59 / 0.89	–	–	–	–
DTU	Pi3X	reprojection	0.041	0.021	0.017	0.12 / 0.91	–	–	–	–
DTU	Pi3X	head, GT labels	0.026	0.021	0.014	0.01 / 0.90	–	0.79 / 0.05 / 0.16	0.006	0.016
DTU	Pi3X	head, label-free	0.026	0.021	0.014	0.02 / 0.90	–	0.77 / 0.06 / 0.17	0.005	0.016
CO3Dv2	VGGT-1B	native confidence	0.105	0.026	0.019	0.83 / 0.90	0.00 / 0.00	–	–	–
CO3Dv2	VGGT-1B	reprojection	0.088	0.026	0.021	1.00 / 1.00	0.14 / 0.53	–	–	–
CO3Dv2	VGGT-1B	head, GT labels	0.068	0.026	0.022	0.17 / 0.82	0.03 / 0.51	0.56 / 0.13 / 0.31	0.014	0.022
CO3Dv2	VGGT-1B	head, label-free	0.070	0.026	0.022	0.16 / 0.82	0.06 / 0.57	0.64 / 0.10 / 0.26	0.034	0.022
CO3Dv2	Pi3X	native confidence	0.122	0.017	0.011	0.80 / 0.84	0.83 / 0.86	–	–	–
CO3Dv2	Pi3X	reprojection	0.065	0.017	0.013	0.18 / 0.89	0.18 / 0.89	–	–	–
CO3Dv2	Pi3X	head, GT labels	0.068	0.017	0.013	0.13 / 0.83	0.06 / 0.71	0.67 / 0.13 / 0.20	0.031	0.014
CO3Dv2	Pi3X	head, label-free	0.067	0.017	0.012	0.12 / 0.84	0.06 / 0.76	0.64 / 0.16 / 0.20	0.019	0.013
ETH3D	VGGT-1B	native confidence	0.196	0.022	0.016	0.68 / 0.89	0.00 / 0.00	–	–	–
ETH3D	VGGT-1B	reprojection	0.125	0.022	0.017	0.03 / 0.93	0.03 / 0.93	–	–	–
ETH3D	VGGT-1B	head, GT labels	0.123	0.022	0.017	0.08 / 0.91	0.00 / 0.54	0.48 / 0.09 / 0.43	0.007	0.013
ETH3D	VGGT-1B	head, label-free	0.123	0.022	0.016	0.07 / 0.90	0.01 / 0.62	0.57 / 0.07 / 0.36	0.008	0.012
ETH3D	Pi3X	native confidence	0.145	0.015	0.009	0.61 / 0.90	0.62 / 0.92	–	–	–
ETH3D	Pi3X	reprojection	0.124	0.015	0.012	0.04 / 0.96	0.04 / 0.98	–	–	–
ETH3D	Pi3X	head, GT labels	0.127	0.015	0.010	0.08 / 0.90	0.01 / 0.66	0.55 / 0.13 / 0.32	0.037	0.009
ETH3D	Pi3X	head, label-free	0.129	0.015	0.010	0.13 / 0.90	0.05 / 0.79	0.60 / 0.11 / 0.29	0.052	0.009
Photot.	VGGT-1B	native confidence	0.030	0.039	0.020	0.57 / 0.89	0.00 / 0.02	–	–	–
Photot.	VGGT-1B	reprojection	0.024	0.039	0.025	0.30 / 0.91	0.30 / 0.90	–	–	–
Photot.	VGGT-1B	head, GT labels	0.013	0.039	0.026	0.27 / 0.90	0.06 / 0.59	0.67 / 0.11 / 0.22	0.007	0.026
Photot.	VGGT-1B	head, label-free	0.012	0.039	0.023	0.20 / 0.90	0.06 / 0.61	0.72 / 0.10 / 0.18	0.007	0.024
Photot.	Pi3X	native confidence	0.020	0.041	0.021	0.63 / 0.88	0.40 / 0.71	–	–	–
Photot.	Pi3X	reprojection	0.025	0.041	0.031	0.31 / 0.90	0.33 / 0.94	–	–	–
Photot.	Pi3X	head, GT labels	0.011	0.041	0.023	0.17 / 0.90	0.13 / 0.83	0.83 / 0.09 / 0.09	0.012	0.028
Photot.	Pi3X	head, label-free	0.011	0.041	0.023	0.17 / 0.90	0.19 / 0.92	0.87 / 0.06 / 0.07	0.015	0.028

Table S9. **Cross-model disagreement as a detector.** Filled patches: DTU seed 0, blur, radius 0.18 (18 scans); CO3Dv2 blur and noise, seed 0, radius 0.18 (18 scenes). Single-view: the DTU vis protocol pooled over 3, 5 and 10 views. 95% cluster-bootstrap intervals over scenes.

pair	filled patch, DTU		filled patch, CO3Dv2		single-view, DTU	
	AUROC	CUR	AUROC	CUR	AUROC	CUR
Pi3X against DA3	0.560 [0.497, 0.624]	0.831 [0.761, 0.892]	0.563 [0.513, 0.614]	0.807 [0.740, 0.869]	–	–
Pi3X against Pi3	0.574 [0.493, 0.651]	0.794 [0.710, 0.873]	0.610 [0.557, 0.660]	0.759 [0.690, 0.821]	0.592 [0.545, 0.636]	0.733 [0.687, 0.776]
VGGT-1B against DA3	0.548 [0.461, 0.631]	0.822 [0.738, 0.893]	0.545 [0.478, 0.607]	0.831 [0.769, 0.888]	–	–
VGGT-1B against MapAnything	0.613 [0.526, 0.698]	0.779 [0.651, 0.892]	0.533 [0.459, 0.606]	0.856 [0.789, 0.910]	–	–
VGGT-1B against Pi3	0.571 [0.496, 0.641]	0.781 [0.703, 0.853]	0.564 [0.494, 0.629]	0.810 [0.745, 0.871]	0.571 [0.510, 0.628]	0.746 [0.684, 0.804]
VGGT-1B against Pi3X	0.523 [0.440, 0.602]	0.865 [0.809, 0.916]	0.527 [0.452, 0.595]	0.842 [0.779, 0.900]	0.543 [0.476, 0.606]	0.765 [0.708, 0.821]

by a per-view median ratio over the pixels outside the hidden region (the input mask, not a label), and every pixel's score is minus the relative spread of its depth over the three fills, so a pixel whose depth depends on the fill scores low. Table S10: AUROC and CUR of that score on the cell's U / S labels, the same for the blur-noise pair alone, the median relative spread inside and outside the hidden region, native confidence on the blur run of the same cells, and the single-view variant (the region refilled in view 0 only, gray and noise against blur, scored on the pixels of that view and on those of the nine untouched views). Refilling the region in a single view moves the surface in that view (AUROC 0.93 to 0.97 on its pixels, median spread 0.7 to 5.4%) and, an order of magnitude less, in the nine untouched views (0.73 to 0.88 on theirs, spread 0.1 to 0.7%): the invented surface follows the pixels of the view it is drawn in, with a weaker pull from the other views. 18 DTU scans and 18 CO3Dv2 scenes, four backbones.

Table S10. **Fill sensitivity of the invented surface.** The depth spread over three fills of the same hidden sphere as a detector, against native confidence on the same cells; the last two columns of each block refill one view only. 95% cluster-bootstrap intervals over scenes.

backbone	DTU, 18 scans								CO3Dv2, 18 scenes							
	AUROC	CUR	AUROC (blur, noise)	spread U (%)	spread S (%)	native AUROC	one view, own	one view, others	AUROC	CUR	AUROC (blur, noise)	spread U (%)	spread S (%)	native AUROC	one view, own	one view, others
VGGT-1B	0.978 [0.965, 0.987]	0.053 [0.028, 0.083]	0.934	2.0	0.1	0.915	0.969	0.882	0.910 [0.871, 0.943]	0.209 [0.135, 0.292]	0.844	1.1	0.2	0.839	0.934	0.843
Pi3X	0.918 [0.889, 0.943]	0.225 [0.152, 0.312]	0.854	1.6	0.3	0.831	0.930	0.842	0.965 [0.952, 0.976]	0.079 [0.049, 0.117]	0.903	1.1	0.1	0.778	0.935	0.823
Pi3	0.927 [0.873, 0.962]	0.174 [0.099, 0.274]	0.876	2.1	0.3	0.928	0.938	0.754	0.947 [0.930, 0.963]	0.125 [0.080, 0.177]	0.887	0.8	0.1	0.934	0.944	0.843
Depth Anything 3	0.824 [0.767, 0.875]	0.437 [0.318, 0.563]	0.806	2.0	0.6	0.714	0.960	0.729	0.620 [0.543, 0.699]	0.723 [0.604, 0.823]	0.612	1.3	0.9	0.684	0.775	0.630

The matched split. Table S11: every detector on the matched-patch protocol of the paper's Section 3 (three identical patches per scene: hidden everywhere, hidden except in an evidence set of 1, 2 or half the views, hidden only in the evidence set), DTU and CO3Dv2, blur and noise pooled, seeds 0 and 1, radius 0.18. AUROC$_m$ and CUR$_m$ score the imagined patch against the filled-but-observed one; *patch* scores clean supported pixels against the filled-but-observed ones; the evidence columns split the observed pixels by how many clean views see them (one at low or high parallax, two or more). The per-image controls are the pipeline check: they must stay near chance.

Table S11. **Matched split, every detector.**

detector	DTU						CO3Dv2					
	AUROC$_m$ ↑	CUR$_m$ ↓	patch	1 view, low	1 view, high	2+ views	AUROC$_m$ ↑	CUR$_m$ ↓	patch	1 view, low	1 view, high	2+ views
VGGT-1B												
native confidence	0.551	0.802	0.909	0.521	0.503	0.499	0.486	0.852	0.679	0.525	0.466	0.473
reprojection, own cameras	0.495	0.903	0.527	0.494	0.485	0.467	0.480	0.909	0.459	0.490	0.459	0.485
support head, label-free (clean-view labels)	0.544	0.780	0.742	0.459	0.466	0.516	0.464	0.893	0.626	0.475	0.432	0.433
matched per-image control (pipeline check)	0.587	0.694	0.999	0.577	0.610	0.581	0.575	0.743	0.987	0.573	0.578	0.590
matched linear probe	0.726	0.504	0.982	0.686	0.691	0.787	0.629	0.703	0.962	0.633	0.592	0.681
matched head, GT placement	0.763	0.430	0.993	0.725	0.727	0.825	0.656	0.616	0.965	0.649	0.648	0.712
matched per-image control, label-free	0.562	0.696	0.999	0.559	0.569	0.558	0.538	0.805	0.981	0.543	0.545	0.559
matched head, label-free	0.754	0.454	0.992	0.707	0.726	0.826	0.681	0.619	0.931	0.685	0.638	0.744
Pi3X												
native confidence	0.474	0.870	0.822	0.499	0.407	0.430	0.500	0.806	0.767	0.536	0.473	0.496
reprojection, own cameras	0.467	0.900	0.520	0.472	0.437	0.439	0.485	0.913	0.483	0.490	0.470	0.476
support head, label-free (clean-view labels)	0.481	0.820	0.693	0.398	0.437	0.452	0.502	0.845	0.481	0.516	0.437	0.475
matched per-image control (pipeline check)	0.591	0.670	0.999	0.585	0.620	0.580	0.583	0.718	0.981	0.586	0.595	0.595
matched linear probe	0.712	0.577	0.948	0.694	0.680	0.779	0.659	0.650	0.886	0.664	0.614	0.723
matched head, GT placement	0.730	0.503	0.985	0.710	0.715	0.808	0.687	0.551	0.925	0.681	0.674	0.754
matched per-image control, label-free	0.614	0.620	0.998	0.614	0.602	0.631	0.565	0.757	0.982	0.559	0.548	0.600
matched head, label-free	0.693	0.537	0.979	0.680	0.658	0.758	0.675	0.644	0.896	0.664	0.627	0.743
Pi3												
native confidence	0.495	0.828	0.925	0.495	0.494	0.484	0.494	0.808	0.905	0.518	0.521	0.492
reprojection, own cameras	0.501	0.891	0.524	0.497	0.484	0.482	0.475	0.924	0.494	0.482	0.462	0.472
matched per-image control (pipeline check)	0.585	0.694	0.999	0.572	0.601	0.579	0.571	0.748	0.988	0.570	0.580	0.588
matched head, GT placement	0.768	0.454	0.983	0.740	0.728	0.859	0.685	0.593	0.949	0.679	0.643	0.761
matched per-image control, label-free	0.592	0.683	0.999	0.578	0.588	0.578	0.570	0.756	0.981	0.578	0.566	0.584
matched head, label-free	0.772	0.448	0.988	0.748	0.729	0.855	0.678	0.627	0.946	0.691	0.616	0.740
Depth Anything 3												
native confidence	0.425	0.854	0.751	0.420	0.414	0.354	0.482	0.839	0.663	0.528	0.479	0.446
reprojection, own cameras	0.431	0.924	0.570	0.429	0.410	0.385	0.484	0.886	0.479	0.488	0.458	0.464
matched per-image control (pipeline check)	0.591	0.773	0.724	0.597	0.607	0.614	0.581	0.777	0.736	0.570	0.572	0.621
matched head, GT placement	0.703	0.548	0.996	0.701	0.678	0.817	0.622	0.654	0.976	0.649	0.624	0.679
matched per-image control, label-free	0.594	0.749	0.711	0.608	0.590	0.601	0.582	0.781	0.727	0.572	0.568	0.616
matched head, label-free	0.714	0.516	0.991	0.712	0.670	0.811	0.671	0.606	0.955	0.679	0.657	0.727

Provenance. Table S12: the provenance heads of Section 5 of the paper on the vis cells of every dataset, support AUROC and CUR of the aggregated score next to the conv head of the main table on the same cells, and the attribution columns (prov-AUROC, its in-frame variant that keeps only views whose frame contains the point, top-1) against the reprojection test with the model's own cameras read per view. Means over scene means. Figure S8 shows one scene per dataset.

Table S12. **Provenance heads.** Support score (AUROC, CUR) and attribution (prov-AUROC over every ordered pair of views, its in-frame variant, top-1 on supported pixels) per backbone and dataset.

detector	DTU					CO3Dv2					ETH3D					Photot.				
	AUROC	CUR	prov-AUROC	in-frame	top-1	AUROC	CUR	prov-AUROC	in-frame	top-1	AUROC	CUR	prov-AUROC	in-frame	top-1	AUROC	CUR	prov-AUROC	in-frame	top-1
VGGT-1B																				
support head, label-free (no attribution)	0.992	0.017	–	–	–	0.931	0.199	–	–	–	0.980	0.051	–	–	–	0.930	0.191	–	–	–
reprojection, own cameras	0.953	0.083	0.965	0.944	0.990	0.792	0.702	0.783	0.783	0.937	0.975	0.069	0.959	0.905	0.827	0.824	0.340	0.889	0.765	0.951
provenance head, GT labels	0.995	0.011	0.971	0.939	0.992	0.924	0.208	0.835	0.831	0.939	0.977	0.058	0.879	0.828	0.779	0.926	0.216	0.884	0.784	0.955
provenance head, label-free	0.993	0.014	0.958	0.921	0.988	0.926	0.195	0.829	0.826	0.940	0.970	0.079	0.828	0.832	0.766	0.932	0.177	0.853	0.776	0.938
Pi3X																				
support head, label-free (no attribution)	0.988	0.025	–	–	–	0.939	0.190	–	–	–	0.963	0.103	–	–	–	0.940	0.166	–	–	–
reprojection, own cameras	0.914	0.164	0.911	0.880	0.981	0.884	0.254	0.912	0.911	0.951	0.977	0.054	0.976	0.923	0.831	0.814	0.361	0.878	0.756	0.950
provenance head, GT labels	0.992	0.018	0.953	0.917	0.986	0.903	0.271	0.870	0.870	0.945	0.965	0.094	0.818	0.840	0.758	0.923	0.201	0.823	0.790	0.906
provenance head, label-free	0.989	0.025	0.943	0.911	0.985	0.913	0.250	0.877	0.877	0.948	0.958	0.115	0.822	0.852	0.752	0.928	0.187	0.829	0.793	0.909
Pi3																				
support head, label-free (no attribution)	0.989	0.021	–	–	–	0.930	0.225	–	–	–	0.951	0.139	–	–	–	0.931	0.181	–	–	–
reprojection, own cameras	0.952	0.089	0.941	0.924	0.987	0.894	0.205	0.930	0.929	0.953	0.976	0.039	0.972	0.937	0.831	0.817	0.345	0.887	0.761	0.950
provenance head, GT labels	0.993	0.016	0.956	0.925	0.987	0.940	0.185	0.877	0.876	0.940	0.961	0.103	0.807	0.837	0.760	0.918	0.223	0.824	0.791	0.915
provenance head, label-free	0.991	0.019	0.951	0.925	0.986	0.947	0.152	0.881	0.879	0.942	0.946	0.150	0.815	0.856	0.762	0.923	0.201	0.820	0.801	0.918
Depth Anything 3																				
support head, label-free (no attribution)	0.982	0.040	–	–	–	0.879	0.323	–	–	–	0.974	0.070	–	–	–	0.925	0.202	–	–	–
reprojection, own cameras	0.926	0.175	0.890	0.870	0.981	0.836	0.463	0.854	0.853	0.949	0.962	0.109	0.934	0.868	0.828	0.844	0.332	0.866	0.755	0.953
provenance head, GT labels	0.985	0.037	0.937	0.895	0.980	0.887	0.295	0.847	0.845	0.938	0.971	0.069	0.842	0.767	0.777	0.922	0.202	0.810	0.761	0.905
provenance head, label-free	0.982	0.044	0.930	0.892	0.980	0.887	0.301	0.850	0.848	0.941	0.966	0.084	0.841	0.798	0.793	0.924	0.202	0.797	0.770	0.902

Twenty views. With 20 clean DTU views, twice the most the heads saw in training, the visibility heads score AUROC 0.990 to 0.993 and the label-free heads 0.990 to 0.992, native confidence falls to 0.61 (VGGT) and 0.83 (Pi3X), reprojection

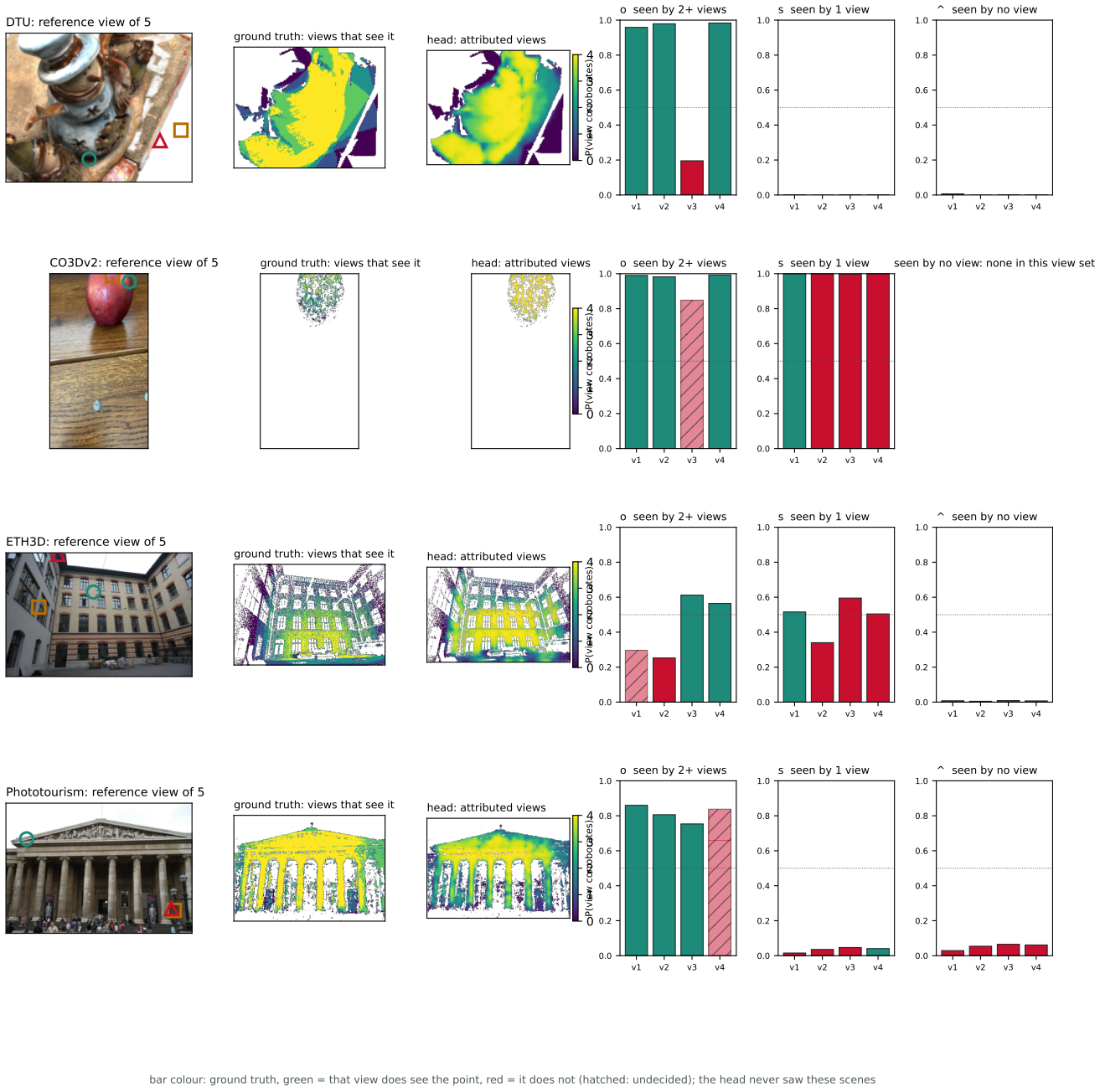


Figure S8. **Which view saw it, one scene per dataset** (VGGT, label-free provenance head, five input views, scenes the head never saw). Per row: the reference view with three query pixels, the ground-truth number of other views that see each pixel, the number the head attributes, and the head's probability per other input view for each query (bar colour: ground truth).

consistency scores 0.93 and 0.89, and the per-image control rises to 0.94: with twenty views the few remaining single-view pixels sit at the frame edge, and the control reads the edge.

Real occlusion against dense references: protocol. The backbone runs on N clean input views (3, 5 or 10; DTU: the spread view sets rotated by the seed; CO3Dv2 and ETH3D: adjacent frames, ETH3D ordered by co-visibility). Every input pixel is back-projected with the predicted depth and cameras and placed in the ground-truth frame by a similarity transform: fitted on the camera centres on DTU and ETH3D, and on the input pixels that carry ground-truth depth on CO3Dv2, whose adjacent orbit frames make a fit on camera centres degenerate (a cell then needs at least half of those pixels on the scanned surface; the measured share is 0.86 to 0.88). Each predicted point is matched to the nearest point of the dense reference (DTU:

all 49 rig views; CO3Dv2: the MVS depth of every known frame; ETH3D: the laser depth of every image), at most 600k points, within $\tau = 1\%$ of the median depth. A reference point is *observed* when at least one input view sees it (it projects into the view onto that view's own ground-truth depth within 3%) and *unobserved* otherwise. Predicted points on an unobserved surface are the positives; the score is the detector's confidence; AUROC is computed per (scene, seed, view count) when at least 30 points are unobserved. A cell (dataset, backbone, view count) supports a detection claim when the unobserved share averages at least 0.5% and the head beats native confidence at $p < 0.05$ (paired sign-flip over scenes).

Validity of the observed test. The observed test needs ground-truth depth in the input view; a hole there labels a surface the camera does see as unobserved. We measure this independently: every reference point is rendered into each input view as a z-buffer (3×3 minimum filter), and the *hole rate* is the share of z-visible points that the depth test calls unobserved. A cell is valid when its mean hole rate is at most 0.10. DTU: 0.001 to 0.004. CO3Dv2: 0.034 to 0.081. ETH3D: 0.31 to 0.35 (0.09 to 0.32 on three scenes when every reference point is rendered instead of 600k), so ETH3D is reported in Table S13 but not used for the claim.

Capacity. Table S14: three heads trained on the same frozen features with the same label-free labels and schedule (1500 steps), scored with the real-occlusion protocol, plus the per-image control. Parameters: VGGT-1B: linear 2.11 M, conv 3.29 M, transformer 3.71 M; Pi3X: linear 0.53 M, conv 1.71 M, transformer 2.12 M The cross-view transformer head has two global-attention blocks over the tokens of all views (2D sinusoidal positions, a learned view embedding). It reaches a validation AUROC of 0.993 on its own labels yet is better than the conv head in none of the 12 cells.

Fused reconstruction at three views. Table S15: the fused-cloud protocol above with three input views, where real unseen geometry is most common, on the 18 DTU benchmark scans, the 22 DTU test scans and 12 ETH3D scenes, two view sets each. The head's cloud is about as accurate as native confidence's and keeps less single-view geometry in every row.

The matched head and the one-view case. On the matched split the per-image controls score 0.585 to 0.591 AUROC_m on DTU (bound 0.60), the heads 0.703 to 0.768 with ground-truth placement and 0.693 to 0.772 label-free. With one clean low-parallax view the heads reach 0.68 to 0.75; fine-tuning the VGGT heads for 1000 more steps with one-view evidence sampled at probability 0.7 leaves that case at 0.713 and 0.715, so the limit is the evidence, not the training budget.

Gaussian-splatting backbone (partial run). AnySplat (reference in the paper), a feed-forward 3D Gaussian-splatting model, was scored on the DTU configuration with its per-pixel Gaussian opacity as confidence (120 of 540 rows) and with the rendered accumulated alpha (450 of 540 rows), bf16 on one L4. Pooled over blur and noise: opacity AUROC 0.583 / CUR 0.718, rendered alpha 0.632 / 0.745, CUR-ERR 0.02 and 0.06; the noise fill drives the opacity below chance (0.36). The representation does not change the answer: a Gaussian the model places on an unseen surface is as opaque as one on an observed surface. The run stopped behind other jobs on the shared card and is reported as partial.

Table S13. **Real occlusion against dense references, every cell.** AUROC for predicted points on observed against unobserved surfaces, mean over scenes (two seeds each); paired sign-flip p over scenes. *claim*: yes / no where the cell qualifies (valid, share $\geq 0.5\%$), – where the share is below 0.5%, invalid where the hole rate exceeds 0.10. DTU benchmark 18 scans, DTU test 22, CO3Dv2 18 scenes, ETH3D 12 scenes.

dataset	backbone	views	hole rate	unobs. share	native	reproj.	head	head − native (p)	claim
DTU benchmark	VGGT-1B	3	0.004	1.35%	0.797	0.757	0.815	+0.017 (0.294)	no
DTU benchmark	VGGT-1B	5	0.002	0.81%	0.797	0.811	0.896	+0.099 (<0.0001)	yes
DTU benchmark	VGGT-1B	10	0.001	0.36%	0.782	0.717	0.918	+0.135 (<0.0001)	–
DTU benchmark	Pi3X	3	0.004	1.09%	0.682	0.628	0.776	+0.094 (0.0003)	yes
DTU benchmark	Pi3X	5	0.002	0.53%	0.696	0.675	0.822	+0.127 (<0.0001)	yes
DTU benchmark	Pi3X	10	0.001	0.17%	0.710	0.600	0.894	+0.184 (<0.0001)	–
DTU benchmark	Pi3	3	0.004	1.14%	0.658	0.651	0.759	+0.101 (0.005)	yes
DTU benchmark	Pi3	5	0.002	0.64%	0.743	0.707	0.849	+0.107 (<0.0001)	yes
DTU benchmark	Pi3	10	0.001	0.20%	0.797	0.662	0.891	+0.094 (<0.0001)	–
DTU benchmark	DA3	3	0.004	1.70%	0.605	0.614	0.707	+0.102 (0.0003)	yes
DTU benchmark	DA3	5	0.002	0.69%	0.675	0.633	0.775	+0.100 (0.0003)	yes
DTU benchmark	DA3	10	0.001	0.23%	0.747	0.618	0.886	+0.139 (0.0001)	–
DTU test	VGGT-1B	3	0.003	1.04%	0.807	0.762	0.849	+0.042 (0.069)	no
DTU test	VGGT-1B	5	0.002	0.68%	0.820	0.811	0.904	+0.084 (<0.0001)	yes
DTU test	VGGT-1B	10	0.001	0.30%	0.806	0.727	0.928	+0.121 (<0.0001)	–
DTU test	Pi3X	3	0.003	0.95%	0.672	0.656	0.753	+0.082 (0.0003)	yes
DTU test	Pi3X	5	0.002	0.53%	0.723	0.675	0.823	+0.100 (<0.0001)	yes
DTU test	Pi3X	10	0.001	0.19%	0.746	0.637	0.912	+0.166 (<0.0001)	–
DTU test	Pi3	3	0.003	1.16%	0.633	0.647	0.744	+0.111 (0.001)	yes
DTU test	Pi3	5	0.002	0.55%	0.727	0.701	0.858	+0.131 (<0.0001)	yes
DTU test	Pi3	10	0.001	0.20%	0.819	0.662	0.908	+0.089 (<0.0001)	–
DTU test	DA3	3	0.003	1.11%	0.598	0.627	0.737	+0.139 (0.0001)	yes
DTU test	DA3	5	0.002	0.56%	0.722	0.671	0.823	+0.102 (<0.0001)	yes
DTU test	DA3	10	0.001	0.17%	0.770	0.629	0.912	+0.141 (<0.0001)	–
CO3Dv2	VGGT-1B	3	0.081	3.74%	0.640	0.515	0.648	+0.008 (0.743)	no
CO3Dv2	VGGT-1B	5	0.056	3.16%	0.660	0.519	0.668	+0.008 (0.749)	no
CO3Dv2	VGGT-1B	10	0.034	2.28%	0.676	0.525	0.675	−0.000 (0.994)	no
CO3Dv2	Pi3X	3	0.081	4.19%	0.532	0.513	0.600	+0.068 (0.077)	no
CO3Dv2	Pi3X	5	0.056	3.29%	0.555	0.523	0.613	+0.058 (0.137)	no
CO3Dv2	Pi3X	10	0.034	2.40%	0.618	0.542	0.638	+0.020 (0.560)	no
CO3Dv2	Pi3	3	0.081	4.07%	0.592	0.514	0.610	+0.018 (0.668)	no
CO3Dv2	Pi3	5	0.056	3.52%	0.621	0.525	0.615	−0.005 (0.901)	no
CO3Dv2	Pi3	10	0.034	2.70%	0.680	0.545	0.637	−0.043 (0.349)	no
CO3Dv2	DA3	3	0.081	3.77%	0.624	0.525	0.575	−0.049 (0.125)	no
CO3Dv2	DA3	5	0.056	3.38%	0.655	0.537	0.587	−0.067 (0.043)	no
CO3Dv2	DA3	10	0.034	2.23%	0.719	0.557	0.605	−0.114 (0.0004)	no
ETH3D	VGGT-1B	3	0.349	6.70%	0.551	0.604	0.616	+0.065 (0.027)	invalid
ETH3D	VGGT-1B	5	0.320	6.69%	0.580	0.603	0.655	+0.076 (0.002)	invalid
ETH3D	VGGT-1B	10	0.309	3.86%	0.604	0.593	0.684	+0.080 (0.004)	invalid
ETH3D	Pi3X	3	0.349	7.99%	0.637	0.629	0.639	+0.002 (0.952)	invalid
ETH3D	Pi3X	5	0.320	6.56%	0.657	0.601	0.660	+0.003 (0.864)	invalid
ETH3D	Pi3X	10	0.309	5.49%	0.643	0.587	0.676	+0.033 (0.222)	invalid
ETH3D	Pi3	3	0.349	5.59%	0.623	0.617	0.638	+0.015 (0.626)	invalid
ETH3D	Pi3	5	0.320	5.87%	0.609	0.619	0.658	+0.049 (0.270)	invalid
ETH3D	Pi3	10	0.309	4.07%	0.655	0.577	0.685	+0.030 (0.152)	invalid
ETH3D	DA3	3	0.349	4.44%	0.449	0.592	0.629	+0.180 (0.004)	invalid
ETH3D	DA3	5	0.320	5.50%	0.521	0.572	0.625	+0.103 (0.013)	invalid
ETH3D	DA3	10	0.309	2.18%	0.527	0.550	0.633	+0.106 (0.018)	invalid

Table S14. **Capacity on real occlusion.** AUROC observed against unobserved points, mean over scenes; last column paired sign-flip over scenes. ETH3D cells carry the validity caveat above; on DTU alone the reading is the same.

backbone	dataset	views	native	per-image control	linear probe	conv head (ours)	transformer head	transformer − conv (p)
VGGT-1B	DTU	3	0.797	0.762	0.839	0.815	0.815	+0.001 (0.826)
VGGT-1B	DTU	5	0.797	0.818	0.895	0.897	0.894	−0.004 (0.013)
VGGT-1B	DTU	10	0.782	0.840	0.913	0.918	0.915	−0.003 (0.179)
VGGT-1B	ETH3D	3	0.550	0.541	0.602	0.616	0.576	−0.041 (0.003)
VGGT-1B	ETH3D	5	0.579	0.544	0.638	0.655	0.637	−0.018 (0.117)
VGGT-1B	ETH3D	10	0.604	0.562	0.675	0.684	0.673	−0.011 (0.167)
Pi3X	DTU	3	0.682	0.714	0.773	0.775	0.763	−0.012 (0.001)
Pi3X	DTU	5	0.697	0.774	0.818	0.821	0.816	−0.005 (0.103)
Pi3X	DTU	10	0.710	0.824	0.895	0.892	0.894	−0.002 (0.483)
Pi3X	ETH3D	3	0.636	0.537	0.634	0.639	0.631	−0.008 (0.164)
Pi3X	ETH3D	5	0.658	0.568	0.659	0.660	0.651	−0.009 (0.100)
Pi3X	ETH3D	10	0.646	0.542	0.670	0.674	0.681	+0.006 (0.591)

Table S15. **Fused point cloud at three views.** Accuracy at the given coverage, the head's accuracy and F1 change against native confidence (paired sign-flip p over scenes), and the single-view share of the kept points.

dataset	backbone	coverage	acc. native	acc. reproj.	acc. head	head − native (p)	F1 head − native	single-view native	single-view head
DTU benchmark	VGGT-1B	0.8	0.597	0.569	0.572	−0.024 (<0.0001)	−0.020	0.154	0.101
DTU benchmark	VGGT-1B	0.9	0.556	0.533	0.545	−0.011 (0.0001)	−0.009	0.156	0.131
DTU benchmark	Pi3X	0.8	0.172	0.166	0.173	+0.001 (0.695)	+0.001	0.140	0.099
DTU benchmark	Pi3X	0.9	0.161	0.156	0.159	−0.001 (0.101)	−0.001	0.150	0.128
DTU benchmark	Pi3	0.8	0.209	0.202	0.206	−0.003 (0.107)	−0.003	0.150	0.092
DTU benchmark	Pi3	0.9	0.192	0.186	0.189	−0.003 (0.001)	−0.003	0.153	0.122
DTU benchmark	DA3	0.8	0.175	0.180	0.186	+0.011 (0.449)	+0.011	0.142	0.114
DTU benchmark	DA3	0.9	0.158	0.166	0.170	+0.012 (0.372)	+0.011	0.152	0.140
DTU test	VGGT-1B	0.8	0.516	0.497	0.498	−0.018 (0.0001)	−0.017	0.160	0.104
DTU test	VGGT-1B	0.9	0.484	0.465	0.471	−0.013 (<0.0001)	−0.012	0.162	0.134
DTU test	Pi3X	0.8	0.185	0.178	0.183	−0.002 (0.063)	−0.002	0.144	0.108
DTU test	Pi3X	0.9	0.169	0.163	0.169	−0.000 (0.639)	−0.000	0.155	0.138
DTU test	Pi3	0.8	0.218	0.210	0.212	−0.006 (0.027)	−0.005	0.153	0.097
DTU test	Pi3	0.9	0.200	0.193	0.196	−0.004 (0.003)	−0.004	0.157	0.128
DTU test	DA3	0.8	0.185	0.194	0.195	+0.010 (0.185)	+0.010	0.147	0.121
DTU test	DA3	0.9	0.169	0.177	0.180	+0.011 (0.118)	+0.010	0.158	0.148
ETH3D	VGGT-1B	0.8	0.196	0.191	0.194	−0.001 (0.757)	−0.002	0.038	0.024
ETH3D	VGGT-1B	0.9	0.186	0.183	0.186	+0.000 (0.963)	−0.001	0.037	0.029
ETH3D	Pi3X	0.8	0.188	0.175	0.178	−0.010 (0.021)	−0.013	0.038	0.026
ETH3D	Pi3X	0.9	0.178	0.169	0.173	−0.004 (0.003)	−0.005	0.038	0.031
ETH3D	Pi3	0.8	0.175	0.158	0.164	−0.012 (0.011)	−0.011	0.039	0.026
ETH3D	Pi3	0.9	0.162	0.151	0.156	−0.006 (0.003)	−0.005	0.038	0.030
ETH3D	DA3	0.8	0.119	0.153	0.159	+0.040 (0.060)	+0.044	0.036	0.025
ETH3D	DA3	0.9	0.110	0.143	0.149	+0.038 (0.035)	+0.043	0.036	0.030